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<<organization>>

Data Management Process

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Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0008	Version 1.0 November 20XX	New documentation	John Doe	Jeremy Doe	Internal

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Document Control

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Version History

Date	Version	Author	Changes
28/09/20XX	0.1	John Doe	New Document – initial draft
01/11/20XX	1.0	John Doe	Issued to Business

Reviewers

Name	Role	Date
Jeremy Doe	Data Management Lead	01/11/20XX

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Background

Types of Data

- The types of data employed with AI are well defined (see ISO/IEC 23053:2023 Framework for Artificial Intelligence (AI) Systems Using Machine Learning (ML) for further details), and can be summarized as follows:
    - **Training data**: used to estimate the parameters of candidate models, i.e., train AI models
    - **Validation data**: used to select the best model according to a performance criterium, fine tuning the hyperparameters of a model
    - **Test data**: used to verify the generalization capability of a model and determine the model's performance on production data, i.e., predict how well a model will perform with real-world data
    - **Production data**: comprises operational data to be used by the model for prediction. The distribution of production data can differ from that of training, validation and test datasets.
- Training, validation and test data can be supplemented with simulated and perturbed data (production data to which noise has been added).

Data Sources

- Data management for AI systems covers the steps required to integrate data from multiple sources. The AI models developed and used by <<organization>> draw data from three sources:
    - **Production**: data that has been fed back for use in training, validation or testing through continuous learning
    - **External**: data obtained from external, third-party providers under contract or license to build the widest possible datasets, such as medical databases (e.g., PubMed and the Cochrane Library) or financial forecasting data (e.g., Bloomberg).
    - **Internal**: internal data from <<organization>> used for the training, validation and testing of models, such as employee data (HR records). Such data is employed not only for internal uses of <<organization's>> models, but also for models used by customers. For example, the People Support system is used internally as part of <<organization's>> HR processes by customers. Both customers' and internal data is used to train, test and validate the model.

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IAMS/0008	Version 1.0 November 20XX	New documentation	John Doe	Jeremy Doe	Internal

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Data Architecture

- Each AI system employs multiple datasets. For example, Healthcare AI had five datasets: the Medical Database, Patient Records, Healthcare Professionals, Consultation Data and Research Data. Separate production, training, validation and test data is maintained for each dataset. In essence, there are four instances of each dataset, containing disjoint data.
- The separate production, training, validation and test instances are disjoint to minimize the risk that a model will 'remember' a specific case as part of validation, for example, which would give a false confirmation of the proper operation of the model.

Tools in Use

- The key data management tools in use are:
  - **Hadoop**: provides a data management framework for handling large volumes of data.
  - **Spark**: provides scalable data analytics for large volumes of data.
  - **ProvValidator**: validates the provenance information for data items.
  - **ProvVis**: visualizes the provenance landscape for data items.
- For detailed guidance on the use of these tools, see the separate operational guides:
  - Hadoop Data Management Guide.
  - Data Analytics using Spark.
  - ProvValidator User Guide.
  - ProvVis User Guide.

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Process

Overview

- The Data Management Process is broken down into a number of discrete, in some cases non-linear tasks. The overall process flow is set out in Diagram 1 below.
- Guidance for each of the steps set out in the process flow is provided below the diagram.

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IAMS/0008	Version 1.0 November 20XX	New documentation	John Doe	Jeremy Doe	Internal

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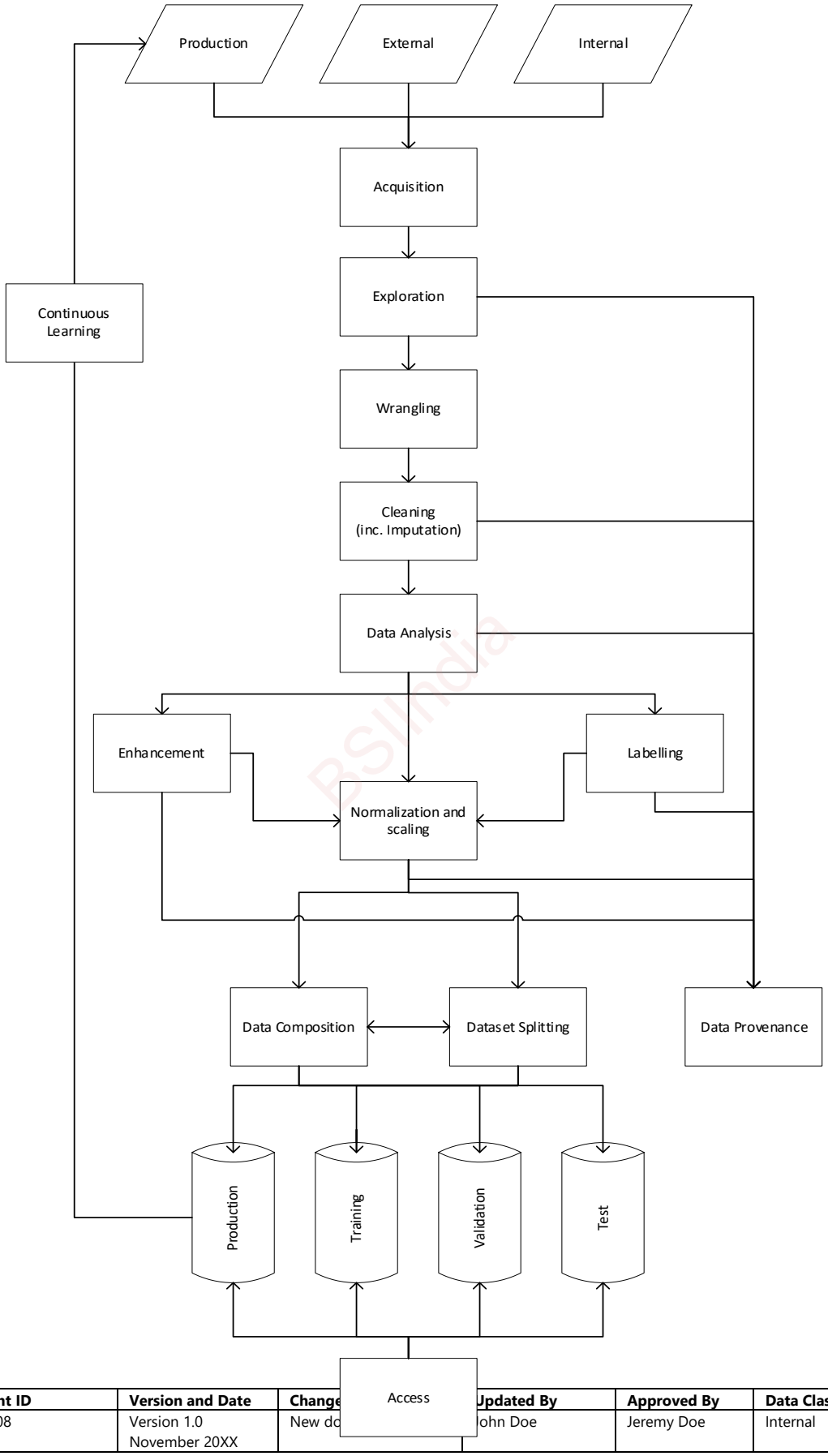

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Diagram 1: Data Management Process Flow

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Process Steps

- Preprocessing steps have been defined in accordance with ISO/IEC 23053:2023.

Data Acquisition

- The mechanism for data acquisition varies depending on the type of data:
  - Production data:
    - Data is fed back from production operations through the continuous learning approach. This is mostly automated, with data delivered through a mixture of real-time feeds and overnight file transfers, depending on the system, details of which are provided in the system specific design documentation.
    - Where systems are operated on customer systems, data fed back is encrypted using a minimum of TLS1.2.
    - Data is collected by the Data Management Team for preprocessing in accordance with the <<Reference>> Data Collection Procedure and stored in the relevant system's data staging area.
  - External data:
    - Data is drawn from external providers in one of:
      - Real-time feeds
      - Overnight feeds, through file transfer
      - Manually, whereby data provided by third parties is entered manually into a database
      - Links to third-party data, similar to real-time feeds but instead of data being fed into an <<organization>> database, data is extracted directly from third-party systems on an as-needed basis.
    - Data is provided by third parties under contract/license, with relationships managed by Procurement.
    - Details of the delivery mechanism for each system are set out in the system specific design documentation.
    - Given data is predominantly in the public domain, feeds are not encrypted unless required by the third party.
    - Data is collected by the Data Management Team for preprocessing in accordance with the <<reference>> Data Collection Procedure and stored in the relevant system's data staging area.

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- Internal data:
  - Internal data is sourced for AI systems that are used within <<organization>>. This is currently limited to:
    - HR AI – a real time feed is provided from the <<organization>> HR system
    - Financial Planning AI – an overnight feed from the <<organization>> general ledger.
  - Given the sensitive nature of the data, feeds are subject to a minimum AES-256 encryption, despite not leaving the <<organization>> network boundary.
  - Data feeds are automated with data placed directly into the relevant system's data staging area, ready for preprocessing.

Exploration

- Data exploration involves the examination of the raw data, to gain insights that help in determining which processes are needed to prepare the data for model training. Depending on the data (and system), the steps undertaken can include:
  - Determine the shape of the dataset, i.e., the distribution of data within the set of data being examined. This can involve examining the data acquired and the potential impact on the dataset into which the data will be inserted.
  - Determine the data type, e.g., text, numeric, image, (Word) document.
  - Statistically analyze the data, i.e., the mean, variance and range of numerical data.
  - Process any provided index information.
  - Determine and process any preexisting features or labels.
- The intelligence gained from exploration can influence which further preprocessing steps are needed for the data. For instance, distribution analysis can help identify outliers, relations and patterns of data.
- Data is drawn from staging areas associated with systems and exploration undertaken in accordance with the <<reference>> Data Exploration Procedure.

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Data Wrangling

- Data wrangling involves the organization of data into well-structured, orderly datasets. Data wrangling processes for tabular data includes aspects such as combining columns, splitting columns, creating columns, changing data types and formats, removing, replacing or adding characters in strings and creating or renaming column headers.
- Data is drawn from staging areas associated with systems and wrangling undertaken in accordance with the <<reference>> Data Wrangling Procedure.

Cleaning

- Data cleaning is undertaken to address duplication and missing data within the sourced data, with the following activities undertaken:
    - Removing duplicate data items.
    - Correcting erroneous data, whereby the data provided is inconsistent with, for example, meta data provided, or failure to conform to an expected data format, such as incorrect postal or zip codes.
    - Identifying missing data or outliers within the data, drawing on the data shape analysis in exploration. The distribution of the data may suggest outliers, or gaps between groups of data or between outliers and the bulk of the data provided.
    - Analyzing missing data to determine whether it should be addressed and if so, in what manner:
        - Assess whether filling in (or indeed correcting) data will impact the validity of the dataset, and whether erroneous data can be discounted or gaps ignored.
        - Assess whether outliers are likely to cause noise in the data, impacting a model's ability to generalize well, and whether outliers can be discounted.
        - Interpolation techniques can be employed to fill gaps in the data.
        - Imputation techniques can be used to replace missing data. Single imputation can be employed if there is no uncertainty about missing values. Alternatively, if there is uncertainty, or multiple imputation can be employed.
- Data is drawn from staging areas associated with systems and exploration undertaken in accordance with the <<reference>> Data Cleaning Procedure.

Data Analysis (analytics)

- Data is analyzed to identify trends and gain insights, for example into the behavior of consumers.

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- This step is distinct from other preprocessing steps in that it does not seek to understand the data for the purposes of training, validation or testing. Instead, it seeks to draw intelligence from the data that can be of use in determining system improvements or new applications for which the data can be used. This helps to influence the future direction of systems development and product marketing.
- Data is drawn from staging areas associated with systems and exploration undertaken in accordance with the <<reference>> Data Analytics Process. Results are fed to relevant department leads depending on the nature of the results obtained.
- It should be noted that the following steps of enhancement and labelling are not required for all data. As part of data analysis, records from data exploration and cleaning are employed to determine whether enhancement and labelling are required, and if not preprocessing continues with normalization and scaling.

Enhancement

- Data enhancement is undertaken for production data only, employing the following approaches:
  - **Enhancement**: refers to the process of improving data with context pulled from additional, external sources, or through the use of simulated data. Data is examined to determine whether enhancement is necessary, and if so, enhancement is undertaken in accordance with the <<reference>> Data Enhancement Procedure.
  - **Perturbation**: this is, predominantly, a data security technique, whereby noise is added to data that obscures 'real' data and enhances data privacy, since individuals' data can be less readily identified. In addition, summary information drawn from the data is not distorted whilst protecting the privacy of individuals referred to within the dataset. Perturbation is undertaken in accordance with the <<reference>> Data Perturbation Procedure, which includes steps to identify bias that may be introduced through the perturbation exercise.

Labelling

- Data labelling is undertaken for training, validation and test data only.
- Data labelling is essentially the addition of meta-data that describes the data, such as a description of the content of an image. However, it always involves the extraction of meaningful information about the data that can describe the data and be used to guide a model's interpretation of the data during training, validation and testing.
- Where meta-data is already present for the data (which if present will have been used as part of data cleaning) the data will be validated and augmented as part of the labelling process.

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- In determining descriptions of the data, the labelling process allows for the identification of personally identifiable information (PII) within the data. It is desirable not to include PII within training, validation and test data, which should be de-identified to preserve the privacy of individuals and minimize the risk of unauthorized access to PII. This activity is undertaken in consultation with the Cybersecurity Team.
- Data labelling and de-identification are largely manual processes for which there are a range of proforma evaluations forms. Forms have been developed for a multitude of different data and are continually being expanded. See the <<SharePoint folder>> for currently available forms.
- Data is drawn from staging areas associated with systems and labelling undertaken in accordance with the <<reference>> Data Labelling Procedure, whilst the additional de-identification steps are undertaken in accordance with the <<reference>> Data De-identification Procedure.

Normalization and Scaling

- Data normalization and scaling feeds off the analyses performed in exploration and cleaning. The intention is to remove significant variations in numerical data so that the results from the model are appropriate, as follows:
    - **Normalization**: scales data samples to a unit norm (e.g. between O and 1).
    - **Scaling**: conversely creates a normal distribution from the individual samples or compresses the samples into a specified range.
- The following normalization techniques are employed:
    - **Min-max normalization**: linear transformation of the data to fit within minimum and maximum values (often zero and one).
    - **Z-score normalization**: data is scaled based on the mean and standard deviation.
    - **Decimal scaling**: moving the decimal point of the attribute values.
- Not all systems require normalization of their data, so these techniques are only employed for specific systems. Where it is required, data is drawn from staging areas associated with systems and normalization (and scaling) undertaken in accordance with the <<reference>> Data Normalization and Scaling Procedure.

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Data Provenance

- Data provenance involves the recording of information that describes the origin, history, and context of a dataset, aiding the understanding of how the data was collected, processed, and transformed, and what assumptions, methods, and sources were employed. Data provenance is essential for ensuring the quality, reliability, and trustworthiness of data.
- Data provenance information is drawn from multiple steps within this process and is determined and recorded per the IAMS/0009 Data Provenance Process.

Data Composition and Dataset Splitting

- Data composition and data splitting are related and complimentary processes that are undertaken iteratively.
- Data composition involves assembling data from the various sources as processed in preceding steps into a single, coherent dataset for training a specific system, ensuring sufficient distribution and representation of the objects for training the intended features.
- Dataset splitting is the process by which available data is split between separate and disjoint training, validation and test datasets for each system. Data that has been augmented through the preceding steps is separated into one of either the training, validation or test datasets, taking into account the necessary distribution of the data required in each dataset.
- In practice, data is assembled into a single (training) dataset through data composition, and then individual data is split out into the validation and test datasets through dataset splitting. The process is then repeated as necessary until the required distribution is obtained in each dataset.
- Note that data is continuously passing through the preceding steps, so a valid outcome for both data composition and dataset splitting is to pause and await further data to filter through the preceding steps.
- Data is drawn from staging areas associated with systems, and data composition and dataset splitting undertaken in accordance with the <<reference>> Data Composition Procedure and <<reference>> Dataset Splitting Procedure, both of which cross reference each other.

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Data Access

- It is necessary to restrict access to sensitive data (for example PII) within datasets. Through the data analysis and data labelling (de-identification) steps, sensitive data within training, validation and test datasets is minimized. Nevertheless, it is necessary to identify sensitive data where it occurs in all datasets, especially production datasets, and ensure appropriate access restrictions are in place.
- Records from data exploration, provenance, cleaning and labelling are employed to determine if sensitive information is present and readily identifiable in a given dataset. If access restrictions are necessary, this is communicated to the Cybersecurity and IT Teams as necessary.
- Data access assessments are undertaken in accordance with the <<reference>> Data Access Assessment Procedure.

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